Project Status Update

Graded

Group

Enrico Vittori
Sabrina Nicacio
Colin Skinner
View or edit group

Total Points

1 / 1 pts

Question 1

Status Update

1 / 1 pt

✔ **+ 1 pt** Correct

– 0.01 pts Minor comment / issue

Nice work validating your project as you go to manage the complexity here. It seems like you're on track to submit a rock solid project.

Question assigned to the following page: 1

Final Project Update: POMDP Framework for Fuel Mode Switching in Orbital Rendezvous and Docking

Sabrina Nicacio snicacio@stanford.edu
Colin Skinner colin.skinner@stanford.edu
Enrico Vittori enrico.vittori@stanford.edu
AA228/CS238, Stanford University

1. Problem

As autonomous servicing and debris-removal missions become more common in orbit, the need for autonomous orbital rendezvous systems increases. We aim to develop a decision-making framework for an autonomous chaser spacecraft docking onto a satellite. The focus of the task is on fuel mode switching: our satellite will have a dual propulsion system, combining high-thrust chemical engines for long range maneuvers with low-thrust cold gas thrusters for short-range adjustments, as commonly seen on many satellites (Scheinman et al., 2019)

2. Progress

At a high level, the project is comprised of **(i)** a simulation of the environment and dynamics, **(ii)** a Kalman Filter, and **(iii)** the POMDP model. Their interface is visualized in Figure 1.

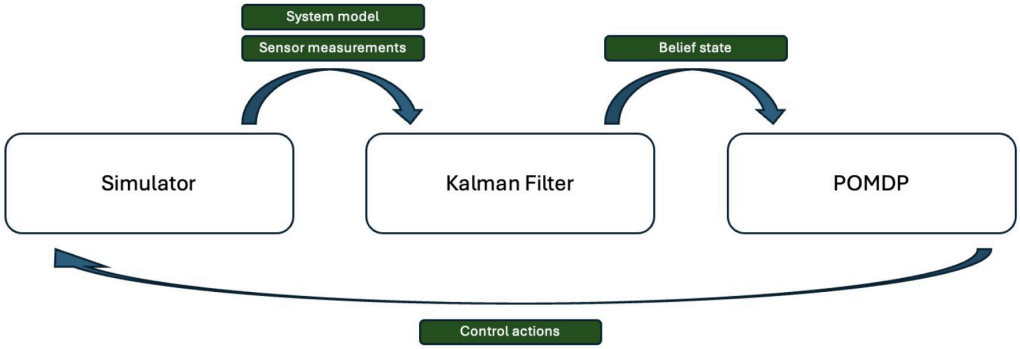


Figure 1: The three modules of the project, including their inputs and outputs

So far, the simulator has been implemented and tested. It employs the Hill-Clohessy-Wiltshire (HCW) equations in discrete time to model the relative translational motion between a chaser and target spacecraft in orbit, given an initial state and control sequence. It generates both the true trajectory and noisy sensor measurements, incorporating process noise (unmodeled disturbances) and measurement noise to create a realistic testbed for the Kalman Filter.

Figure 2 confirms the simulation is working, showing random noise with zero mean and expected variance of $\sigma_p = 0.05m$.

Question assigned to the following page: 1

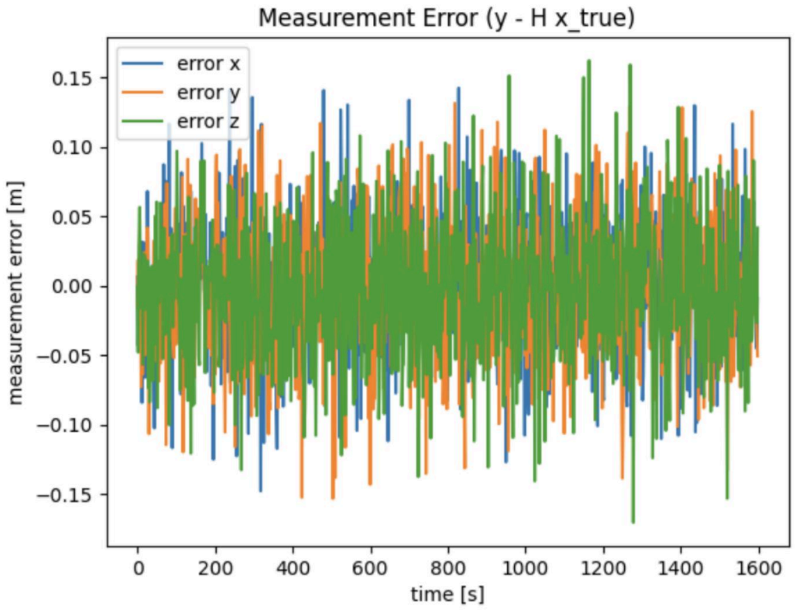


Figure 2: Sensor position measurement error in x-, y-, z- axes

3. Timeline

As we've completed the dynamics model, we're now moving into implementing the Kalman Filter and the POMDP decision-making framework. Our goal is to have both running before Thanksgiving break so we have enough time to test everything and decide whether adding attitude control is feasible. If the base pipeline (dynamics → KF → POMDP) is stable, we'll move on to integrating attitude control; if not, we'll document it as future work. After that, we'll shift our focus to writing the final paper and polishing results.

References

David A. Scheinman et al. Hybrid propulsion systems for spacecraft: Trends and mission applications. *Acta Astronautica*, 160:456–467, 2019.